

Kavya Kolamkar

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Summary

Machine Learning Engineer with hands-on experience building end-to-end geospatial AI pipelines, deep learning models, and full-stack production applications. Skilled in computer vision, retrieval-augmented generation (RAG), and ML deployment using Python, TensorFlow, PyTorch, FastAPI, and React, with proven results across healthcare, agriculture, and document-intelligence domains.

Professional Experience

AI/ML Intern *Forest by HeartfulNess* **Pune, India**
Jan 2026 – May 2026

- Architected an end-to-end geospatial AI pipeline for automated tree species identification using high-resolution drone orthoimagery (GeoTIFF, ≤ 3.5 cm/pixel).
- Engineered a processing pipeline for 4-band RGBA rasters (3–31M pixels per tile), enabling classification across 27+ tree species over agricultural plots in Hyderabad, India.
- Validated the model against 3,300+ GPS-tagged, georeferenced ground-truth trees in UTM Zone 44N, delivering the full deep learning workflow from raw drone imagery to per-tree species predictions.

Technical Skills

Languages & Databases: Python, C/C++, JavaScript, SQL (PostgreSQL, MySQL), MongoDB
LLM & Agentic AI: LangChain, LangGraph, CrewAI, Multi-Agent Systems, Ollama, ChromaDB, RAG
Frameworks & Libraries: React, Node.js, FastAPI, Flask, Django, TensorFlow, PyTorch
Machine Learning: Data Preprocessing, Feature Engineering, CNNs, RNNs, GANs, Supervised & Unsupervised Learning
Deployment & Cloud: Docker, AWS, Vercel, Railway
Data Structures & Algorithms: Dynamic Programming, Graph Algorithms, Trees, Greedy Algorithms, Sliding Window, Hashing, Bit Manipulation

Projects

DermScan – Clinical Skin Lesion Screening & Growth Tracking **Jan 2026**
Python, React, TensorFlow, FastAPI, PostgreSQL, Docker

- Developed a full-stack clinical AI application for early-stage skin lesion screening, training an EfficientNetB3 model on 25,331 ISIC 2019 dermoscopy images.
- Implemented Grad-CAM explainability and automated ABCDE clinical scoring for interpretable predictions, plus longitudinal lesion growth tracking across patient visits.
- Built a full-stack platform with FastAPI and React, including JWT authentication, PostgreSQL persistence, automated PDF report generation, and a Recharts analytics dashboard.
- Containerized the stack with Docker Compose, orchestrating a FastAPI backend, Nginx-served React + Vite frontend, and PostgreSQL database for streamlined deployment.

DocuChat AI – RAG-based Document Intelligence Platform **Mar 2026**
Python, FastAPI, React, LangChain, FAISS, Sentence Transformers

- Built a full-stack RAG application enabling users to upload PDFs and chat with documents via context-aware AI with page-level source citations.
- Designed an end-to-end RAG pipeline using PyMuPDF, LangChain, Sentence Transformers, FAISS, and the Gemini API for document parsing, semantic retrieval, and grounded generation.
- Developed an AI-powered quiz engine generating MCQ, True/False, and short-answer questions with automated evaluation and detailed feedback reporting.
- Architected modular REST APIs for document processing, vector search, chat, quiz generation, and evaluation, reducing hallucination via retrieval-grounded responses with exact page citations.

Education

Indian Institute of Technology, Madras – B.S. in Data Science and Applications (CGPA: 7.0/10) **Jan 2024 – 2028**
International Institute of Information Technology – B.E. in Computer Engineering (CGPA: 8.5/10) **Aug 2023 – 2027**
St. Ursula's High School – Secondary School Certificate (85.00%) **2009 – 2021**

Achievements

CESA Techspark 2024 – 2nd Runner-Up *Team Tri-Netra* **2024**
GDG 2026 – Finalist *Team Slime* **2026**

- Built an AI-agent system for government rainwater-harvesting challenges; owned database design and core agent development.
- Built an AI model for skin disease/cancer classification; led backend development and model implementation.

Leadership & Languages

Social Media Core Team Member *CESA, I2IT* **2025 – 2026**
Languages: English, Marathi, Hindi